

CLEARING THE AIR

The estate sector of Sri Lanka has existed for centuries. Introduced during the British colonial era in the central highlands, it still serves as a key source of export revenue for the country. Sri Lanka is renowned across the globe for its famous teas, but very little has improved for the estate workers, whose hard labour earns the country this much-needed revenue and prestige.

The 'line rooms' serve as a testament to these harsh living conditions and the cycle of generational poverty these communities struggle to escape. 'Line rooms' are small, often a one-room home that houses multiple generations, lacking running water, proper sanitation, and even ventilation. In these living quarters, there is barely enough room to cook, and meals are prepared over firewood hearths.



"I work in this estate, and for cooking, we used to collect firewood from the estate, but during the rainy season, it is very difficult to find firewood. Even when we do, we face problems collecting it from the estate."

-V. Puvuneshuary

For millions of households across Sri Lanka, especially in rural and estate areas, firewood remains the primary cooking fuel. According to the 2012 study 'A Profile of Biomass Stove Use in Sri Lanka' published in the International Journal of Environmental Research and Public Health, biomass fuel use is the leading contributor to indoor air pollution, which disproportionately affects women and children who spend the most time near the hearth.

According to the World Health Organization, household air pollution from burning solid fuels such as firewood causes millions of premature deaths globally each year, with women facing the greatest health burden due to prolonged exposure during cooking. Smoke from inefficient stoves is linked to chronic respiratory disease, heart disease, stroke, lung cancer, and childhood pneumonia. In poorly ventilated dwellings, indoor smoke can reach levels many times higher than safe limits. In estate "line rooms," where kitchens are often part of the living space, the impact is intensified.

"The biggest problem was the smoke when cooking with firewood, because of this smoke, we became sick often."

-V. Puvuneshuary

The same 2012 study further states that women exposed to indoor smoke are significantly more likely to suffer from chronic respiratory illnesses, and that indoor air pollution remains a major but under-addressed public health issue.

Firewood use also comes with a heavy time burden. Collecting fuel, drying it, and managing inefficient cooking takes hours each day, time that estate women do not have. Women in the estate sector already work long, physically demanding hours in plantations under difficult conditions. Firewood collection and slow cooking add another layer of unpaid labour, deepening time poverty. As alternatives, some families turn to LP gas, which comes with its own challenges.



“To buy an LP gas cylinder, we have to travel far into town, the transport costs nearly ten thousand rupees for this.”

The ‘Empowering Women Smallholders Farmers through Renewable Energy-Based Agriculture Modernization Technologies’ project which is funded by the Japanese Supplementary Budget (JSB) of the Government of Japan, has provided 200 estate sector households in Nuwara Eliya with improved, low-cost cookstoves — a modest intervention with far-reaching impact.

The stoves are designed to reduce smoke exposure, use fuel more efficiently, lower costs, reduce health risks, and to cut cooking time.

“After receiving the stove, we can cook quickly and easily, we can feed our children without delay and go to work on time. We also get sick less because of the smoke.”

-V. Parameshuary

Human security is about protecting people from critical threats while enabling them to live with dignity and stability. In Sri Lanka’s estate sector, where multidimensional vulnerability is high, and where energy poverty intersects with health, gender inequality, and economic insecurity, clean cooking solutions address a foundational risk.

Estate field officers have seen the difference.



“After receiving these stoves, the benefits are clear, they cook faster, spend less, can finish their household work and come to work on time.”

-Suntharalingam, estate field officer.



In a country where energy poverty affects millions, and where lack of access to clean cooking fuels is a key driver of inequality, interventions like improved cookstoves do more than reduce smoke. They protect women’s health, save time, stabilize livelihoods, and safeguard human capital, especially in communities already carrying the weight of multiple crises.



In Sri Lanka’s estate sector, development is not only about what happens in the fields. Sometimes, it begins with clearing the air inside the kitchen.